

(12marks)

Q4: Consider the initial value problem:


$$\frac{dy}{dx} = y - x^2 + 1,$$

$$y(0) = 0.5$$

- Solve numerically using Euler's Method with step size $h = 0.2$ for $x = 0.4$.
- Apply the Runge-Kutta Method (RK4) with the same step size for $x = 0.4$.

Q2: You are given the following data points: $(0,1), (1,2), (3,10)$

- Construct the Lagrange interpolating polynomial $P(x)$ that passes through these points. (12marks)
- Use your polynomial to estimate the value of the derivative $f'(2)$.

1: A. For the nonlinear function $f(x) = e^{-x} - x$, use the Taylor series expansion about $x_0 = 0$ up to the fourth order term to approximate $f(2)$. Then, calculate the true percent relative error.

B. Using Gauss Elimination, determine the value of the parameter λ such that the following system of equations has an infinite number of solutions: (12 marks)

p 2020

$$\begin{cases} 2x + y - z = 8 \\ -3x - y + 2z = -11 \\ -2x + y + \lambda z = -4 \end{cases}$$

Question 4:

Given the following data:

x	1	2	3	4	5	6
f(x)	-2	1	10	29	62	113

1. Construct the Newton's forward difference interpolation polynomial.
2. Use the polynomial to find the approximate value of the **first derivative**, $f'(1)$, at the first data point.
3. Use the polynomial to find the approximate value of the **second derivative**, $f''(3)$, at an interior point.
4. Use the polynomial to find the approximate value of the **third derivative**, $f'''(4)$, at another interior point.

Question 5:

Given the differential equation $\frac{dy}{dx} = 2x + y$, with the initial condition $y(0) = 1$.

Using **a single step** of the **RK4 method**, find the approximate solution at $x=0.2$ with a step size of $h=0.2$.

Question 6:

x	1	2	3	4	5	6	7	8
f(x)	2	5	10	17	26	37	50	65

Use numerical integration to find the approximate value of the definite integral $\int_0^7 f(x) dx$ using:

1. Simpson's 1/3 Rule
2. Simpson's 3/8 Rule

Read questions carefully and answer, showing all steps of your solution

Final Answers with no clear solution get ZERO

Question 1 :

Use central difference approximations to estimate the second derivative $O(h^2)$ of $y=e^x \cos x$, at $x=\pi/4$, using a step size $h=\pi/12$.

Question 2:

Use the **best numerical differentiation approximation** to find the rocket's **velocity** and **acceleration** at $t=25$ seconds and $t=125$ seconds.

The data provided is:

t (s)	25	50	75	100	125
y (km)	32	58	78	92	100

Question 3:

Given the following system of linear equations:

$$x + 2y - z = 4$$

$$2x + 4y + 3z = 5$$

$$-x - 2y + kz = -1$$

Find the value of k for which the system of equations has:

1. No solution.
2. An infinite number of solutions.
3. A unique solution.

Final answers with no solution will get ZERO

Q1- - Given:

$$\begin{aligned}2x - y - z &= 2 \\ x - 2y + z &= -4 \\ x + y + \lambda z &= 4\end{aligned}$$

find the value of λ for which the system of equations has no solution. Use Gauss elimination method (12 marks)

Q2: a- Given that $2 \frac{dy}{dx} + y \cos y = 0$, $y(\pi) = \pi$

Using a single step of RK4 method, find the approximate solution at $x = \frac{3\pi}{2}$ (10 marks)

Q3: Use Newton forward-difference formula to construct interpolating polynomials of degree three for the following data . $f(0) = 1$, $f(0.25) = 1.648$, $f(0.5) = 2.718$, $f(0.75) = 4.481$ (8marks)

Q4- For the following data given in the table below for a curved line function $f(x)$

x:	-1	0	1	2	3
f(x)	3.137	6.132	4.311	7.525	2.289

a- Find the area bounded by the curve and x axis from $x = -1$ to $x = 3$ using Simpson's 1/3 rule. (8 marks)

Solution:

b- Use Romberg's method with composite trapezoidal rule to have a better approximation for the area bound by the curve and x axis from $x = -1$ to $x = 3$, take $n = 1, 2, 4$ (12 marks)

SHOW YOUR DETAILED SOLUTIONS AND CALCULATIONS - Final answers

with no solution will get ZERO

Carry your intermediate computations to at least three decimal places.

Q1- Use the golden-section search to find the minimum of $y = x^2 - 4\sin(5x)$
assume that $x_l = 1.3$ and $x_u = 1.7$ (10 marks)

Answer:

Q2- Solve the integral $\int_0^1 \frac{3}{1+x^2} dx$ by Trapezoidal rule with $n = 1, 2, 4$ and 8. Then obtain the better estimate using Romberg integration method. (13 marks)

Answer:

Q3- Consider the ordinary differential equation: $2 \frac{dy}{dx} - 3e^x + 2y = 0$
Given: $y(0) = 0, h = 0.5$. Use the Runge-Kutta method to the 4th order to find $y(1)$.
(12 marks)

Answer:

Q4: From the following data shown in the table below (15 marks)

x:	-1	0	1	3	5
f(x)	3.1	0	4.3	7.5	2.2

1- Find $f(2)$ - employing Newton's forward interpolation formula.

Answer:

2- Find the area bounded by the curve and x axis from $x = -1$ to $x = 5$ using Simpson's 1/3 rule

Answer:

3- Find $f'(3)$ -using two points Centered finite-divided difference formula taking the data from the table.

Answer:

End of questions

The value of an integral $\int_a^b f(x)dx$ given using 1, 2, and 4 segments trapezoidal rule is given as 6.3460, 2.4468, and 1.5536, respectively. The best estimate of the integral you can find using Romberg integration is most nearly

Answer:

Given $4\frac{dy}{dx} + 5y^2 = \sin x$, $y(0.2) = 9$
and using a step size of $h = 0.3$, the value of $y(0.8)$ using Euler's method is most nearly

Answer:

Given $2\frac{dy}{dx} + y^2 = e^x$, $y(0.6) = 8$, and using a step size of $h = 0.3$, the best estimate of $\frac{dy}{dx}(0.9)$ Runge-Kutta 4th order method is most nearly:

Answer:

Q2-Determine the minimum of the function $f(x,y) = x^2 + y^2$. Use point (2, 1) as the initial estimate of the optimal solution. Conduct one iteration. (12marks)

Q3-Evaluate

$$I = \int_2^{2.6} \int_4^{4.9} \frac{1}{xy} dx dy$$

Take $dx = 0.2$ in x-direction, $dy = 0.3$ in y-direction. Use Trapezoidal rule in x-direction and Simpson's 3/8 rule in y-direction.

(10marks)

Q4-Determine the values of a and b for which the system (10marks)

$$\begin{bmatrix} 3 & 0 & 0 \\ 9 & 3 & 2 \\ 1 & 1 & a \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b \\ 3 \\ -1 \end{bmatrix}$$

1) has a unique solution

2) has no solution

and 3) has infinitely many solutions

Question 7 :

Determine the Gauss Quadrature Formula of the following integral:

$$\int_{-1}^1 (2x^2 + 5) dx$$

Use the provided nodes $x_0 = -1$, $x_1 = 0$, $x_2 = 1$.

Question 8 :

If we fit a 2nd order polynomial for 8 values, the correlation coefficient is 0.9990, $\sum (O_i - \hat{y})^2$ is 2500 calculate :

1. The Standard error of estimate based on regression polynomial
2. Standard deviation.

Question 9 :

Given the function:

$$f(x, y) = 2x^2 + 5y^2 + 4xy - 3x - 2y$$

Use the Optimal Steepest Ascent Method to maximize this function, starting from the point (1, -1). Determine the next point.

1- If A is a 4×4 invertible matrix then what is the inverse of $3A$?

- (A) $\frac{1}{4} A^{-1}$ (B) $3 A^{-1}$ (C) $\frac{1}{3} A^{-1}$ (D) $4 A^{-1}$

2- The steepest ascent gradient direction of the function $f(x,y) = y^2 x^2$ at point $(1,2)$ is:

- (A) $16i+12j$ (B) $8i+4j$ (C) $12i+16j$ (D) $4i+8j$

3- Consider the following system of linear equation $3x+y-z=12$, $x+5y+2z=14$, $x+4y+z=9$. The current approximation is $x=4$, $y=2$, $z=0.889$, then the Gauss-Seidal method will give next approximation as $x=3.63$, $y=1.718$ and $z=$

- (A) 1.055 (B) 1.014 (C) 0.930 (D) 0.844

4- The value of k_1 by Runge-Kutta method of fourth order if $dy/dx = x + y^2$ and $y(0.2) = 2$ is:

- (A) 1.131 (B) 0.566 (C) 0.469 (D) 0.2

5- To find the minimum of the function $f(x) = \sin(x) + x^2$ by golden-section search, assume the first interval is $x_l = -0.8$, $x_u = 0$, then the next interval becomes:

- (A) $x_l = 0$, $x_u = -0.4944$ (B) $x_l = -0.8$, $x_u = -0.3056$
(C) $x_l = -0.3056$, $x_u = 0$ (D) $x_l = -0.8$, $x_u = -0.4944$

6- Find the extreme value of y from the following data using Newton's forward interpolation.

la.

x	2	3	5
$f(x)$	5	-5	-7

7- Use the second order Romberg integration formula to compute the following

$$\int_2^8 \frac{9\sin(x)}{x} dx$$

Q1-Answer the following questions (18marks)

The determinant of hessian of the function $x^2 - 2y^2 - 4y + 6$ at point $(0, 0)$ is

Answer:

Given $f(4) = 3, f'(4) = 8, f''(4) = 11$, and that all other higher order derivatives of $f(x)$ are zero at $x = 4$, and assuming the function and all its derivatives exist and are continuous between $x = 4$ and $x = 5$, the value of $f(5)$ is

Answer:

The root of the equation $f(x) = 0$ is found by using the Newton-Raphson method. The initial estimate of the root is $x_0 = 3, f(3) = 5$. The angle the line tangent to the function $f(x)$ makes at $x = 3$ is 57° with respect to the x-axis. The next estimate of the root, x_1 most nearly is

Answer:

Given

$$[A] = \begin{bmatrix} 12.3 & -12.3 & 20.3 \\ 11.3 & -9 & -11.3 \\ 10.3 & -7 & 8.5 \end{bmatrix}, [B] = \begin{bmatrix} 2 & 4 \\ -5 & 6 \\ 12 & -20 \end{bmatrix} \text{ then if } [C] = [A][B], \text{ then } C_{31} =$$

Answer:

The lower triangular matrix $[L]$ in the $[L][U]$ decomposition of matrix given below

$$\begin{bmatrix} 25 & 5 & 4 \\ 10 & 8 & 12 \\ 8 & 12 & 22 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \text{ is}$$

Answer:

The following x, y data is given.

x	6	9	14
y	24	37	25

The Newton's divided difference second order polynomial for the data is given by $f_2(x) = b_0 + b_1(x - 6) + b_2(x - 6)(x - 9)$ The value of b_1 is

Answer:

The following

x	1	2	3	4
y	1	200	400	500

data is regressed with least squares regression to $y = a_0 + a_1x$. The value of a_1 most nearly is

Answer:

The value of $\int_{0.4}^{2.0} e^x dx$ by using 4-segment Simpson's 1/3 rule most nearly is

Answer:

The value of an integral 2.4408, and 1.5536, resp Romberg integration is

Answer:

Given $4 \frac{dy}{dx} + 5y^2 =$ and using a step

Answer:

Given $2 \frac{dy}{dx} + y^2 = e^x$, Runge-Kutta 4th order m

Answer:

Q2-Determine the mi estimate of the optima

Q3- Evaluate

Take $dx = 0.2$ in x-direction 3/8 rule in y-direction.

(10marks)

Q4-Determine the values

1) has a unique solution

Exam Time: 90 minutes
SHOW YOUR DETAILED SOLUTIONS AND CALCULATIONS - Final answers with no solution will not be marked

Answer the following questions

Q1- Fit a straight-line to the data provided in the following Table:

x_i	5	7	9	11	13
y_i	6	4.5	3.4	2.3	1.6

(10 marks)

The equation of the line is:

The value of coefficient of determination, r^2 is:

Q2- Compute the integration: $\int_2^4 \frac{1}{(x-1)^2} dx$, using Work to five significant figures

(10 marks)

Apply Romberg integration taking $n=1, n=2$ & $n=4$ up to $R(3, 2)$.

Answer:

Q3- Evaluate the following double integration: $\int_1^3 \int_1^3 (2x + y) dx dy$.

(10 marks)

Take $dx = 0.5$ in the x -direction, $dy = 0.5$ in the y -direction. Apply the Trapezoidal rule in the x -direction and in the y -direction.

Answer:

Q4- Maximize the function: $f(x, y) = xy + 4x - x^2 - y^2 - 4$

(10 marks)

Apply the Optimal steepest ascent method, use point (2, 2) as the initial estimate of the optimal solution. The optimum point after conducting one iteration is:

Answer:

Q5- Consider the ordinary differential equation: $\frac{dy}{dx} = x^2 - 2y^2 + x$

(10 marks)

Given: $x_0 = 1, y_0 = 2, h = 0.5$

Use the Runge-Kutta method to the 4th order to find $y(1.5)$.

Answer:

End of Questions

Q1-Read the questions carefully and choose one correct option:

- 1- Using a step size of $h = 0.3$ to solve the differential equation $\frac{dy}{dx} + 5y^2 = \sin x$, $y(0.3) = 5$, then the value $y(0.6)$ using Euler's method is most nearly: (34 marks)
 - (A) -32.411
 - (B) 4.577
 - (C) 3.589
 - (D) -48.122
- 2- Applying Newton-Raphson method to estimate the root of $\sqrt{7}$ up to two decimal places. Starting with the initial guess $x_0 = 2$, the first iteration x_1 is:
 - (A) 1.25
 - (B) 2.88
 - (C) 2.75
 - (D) 2.67
- 3- The relative error in an approximate solution is 0.004%. The number of significant digits in the solution that we can trust at least is:
 - (A) 5
 - (B) 3
 - (C) 4
 - (D) 2
- 4- We are given a system of equations containing a real parameter k : $x^2 + y^2 = 12$ and $2x + ky = 22$, then the system has no solution when:
 - (A) $k = 8$
 - (B) $k = 4$
 - (C) $k = 2$
 - (D) $k = 12$
- 5- The value of the double integral using Trapezoidal rule $\int_2^4 \int_2^4 \frac{x}{y^2} dx dy$ is: (take the step size $h=1, k=1$)
 - (A) 0.469
 - (B) 0.154
 - (C) 0.108
 - (D) 0.118
- 6- Which of the following is a suitable initial interval for computing a positive root of the equation using the bisection method, given that: $g(x) = \cos(x) - 0.5e^x$
 - (A) $[-1, 0.1]$
 - (B) $[-1, 0.2]$
 - (C) $[0.1, 0.4]$
 - (D) $[0.4, 0.7]$
- 7- If the function studied by bisection method changes sign between the initial values 3 and 4 and we know that an error less than 0.00583 is acceptable. Then the required number of iterations that will produce the desired result is:
 - (A) 8
 - (B) 12
 - (C) 14
 - (D) 16
- 8- The four-point central difference approximation of the first derivative of the $f(x) = \frac{x}{x-1}$ at $x = 3$ and the step size $h = 1$ is:
 - (A) 2.767
 - (B) 0.967
 - (C) 1.433
 - (D) 1.800
- 9- The value of $\int_{8.1}^{8.5} x \cdot \sin(x) dx$ using 1/3 Simpson Rule with $h = 0.2$ is:
 - (A) 2.973
 - (B) 9.873
 - (C) 7.377
 - (D) 4.100
- 10- If (0.333) is approximated by a second Lagrange Polynomial for $f(x) = 2^x$ through the points 0, 1, 2, then the approximate value $f(0.333)$ is:
 - (A) 1.222
 - (B) 1.275
 - (C) 1.249
 - (D) 1.200